
MA 641: Operations Research

Course Code	MA 641
Course Name	Operations Research
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	Ph.D./ M.S./B.Tech
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Module I** [10 Lectures]

Introduction to Operations Research, Models and Modelling in Operations Research, Graphical Method, Simplex Method and its variants.

- **Module II** [12 lectures]

Sensitivity Analysis, Duality and Post-Optimal Analysis, Advanced Linear Programming: Bounded Variables, Parametric Linear Programming, Revised Simplex Algorithm, Goal Programming, Dual Simplex Method, Integer Linear Programming.

- **Module III** [12 lectures]

Transportation Model and its Variants: Balanced and Unbalanced Transportation Problem, Transshipment, Assignment Problem: Auction and Hungarian Method, unbalanced assignment problem. Sequencing Problem and Variants: Algorithms for processing n-jobs through m-machines. Traveling Salesman Problem, Heuristics and Branch and Bound and Gomory's Algorithms.

- **Module IV** [6 lectures]

Project Evaluation and Review Technique, Critical Path Method.

Text and Reference Books

1. Saul I. Gass, **Linear Programming: Methods and Applications**, Dover publications, 2010.
2. Hamdy A. Taha, **Operations Research: An Introduction**, Pearson Education, 2008.
3. Don T. Phillips, A. Ravindran, James J. Solberg, **Operations Research: Principles and Practice**, John Wiley & Sons, 1987.
4. George Bernard Dantzig, **Linear Programming: Theory and extensions**, Princeton University Press, 1998.
5. G. Hadley, **Linear Programming**, Addison-Wesley, 1962.
6. Michael W. Carter, Camille C. Price, Camille C. Price, **Operations Research**, CRC Press, 2000.
7. Frederick S. Hillier, Gerald J. Lieberman, **Introduction to operations research**, McGraw-Hill, 2001.